What is a network?

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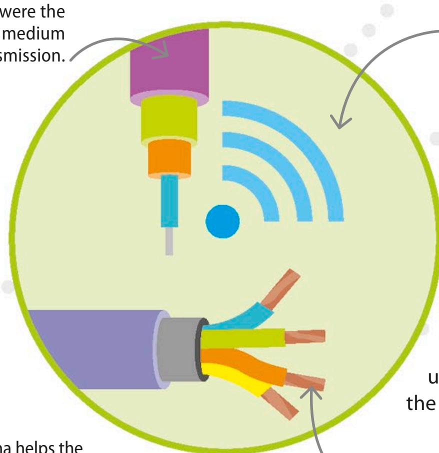
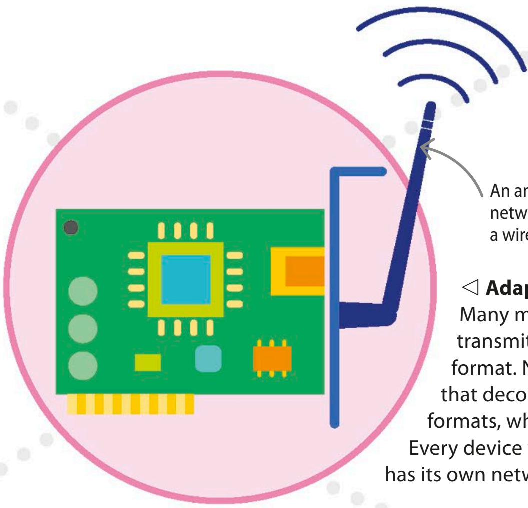
A network is a group of connected devices, and can include computers, smartphones, printers, routers, and hard drives. Its purpose is to share resources and data.

How does it work?

A network node is any device that sends or receives data through the network. Designed and created as per the needs of its users, networks can be big or small, public or private, and can have varying levels of security. The internet, for instance, is a massive public network that spans the entire world.



◁ **Connectors**
A medium is needed to transmit signals between two nodes. Copper wires are a common connector, as are wireless radio waves (such as Wi-Fi, 3G, and 4G). Cell towers, satellites, and undersea cables are all part of the communication infrastructure.

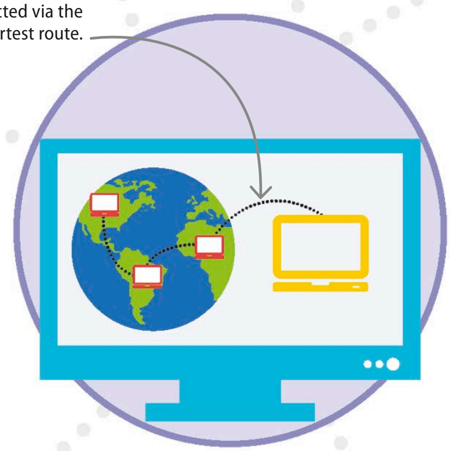


◁ **Adapters**
Many media, such as telephone wires, transmit information in an analogue format. Network adapters are hardware that decode analogue signals into digital formats, which the computer can read. Every device with an internet connection has its own network adapter.

LINGO

Communication protocols

When two devices communicate, a protocol dictates whose turn it is to send data, what kind of data is being sent, and how this data is formatted.
Protocol: A set of rules that governs the transmission of data between devices.
HTTP (Hypertext Transfer Protocol): Used for visiting webpages.
HTTPS (Hypertext Tranfer Protocol Secure): A secure HTTP.
DHCP (Dynamic Host Configuration Protocol): All computers use this to obtain their IP address from a router.



▷ **Routing**
A computer in one part of the world can't be wired to a device in another. Communication between computers over long distances involves hopping from node to node until the target is reached. Finding the shortest path between two devices is called routing.